A SIMPLE PROOF OF THE ERDŐS–SÓS CONJECTURE

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Abstract. We give a simple version of the proof obtained by OpenAI [2, 3] for the Erdős–Sós Conjecture [1]: for any positive integers n and t, every graph with n vertices and more than $(t-2)n/2$ edges contains a copy of every tree with t vertices.

1. A proof of the Erdős–Sós Conjecture

A *rooted tree* (T, x) is a tree T and a distinguished vertex $x \in V(T)$. Given an n-vertex graph G, a copy of (T, x) in G *rooted* at $u \in V(G)$ is a copy of T in G with x mapped to u. For any tree U and $x \in V(U)$, denote by $\mathcal{B}_G(U, x)$ the family of pairs (σ, v_j) where $\sigma = (v_1, \dots, v_n)$ is an ordering of $V(G)$ and $v_j \in N_G(v_1)$ such that $G[\{v_1, \dots, v_{j-1}\}]$ contains no copy of (U, x) rooted at v_1.

Proof of the Erdős–Sós Conjecture. Let G be an n-vertex graph with no copy of a tree T on $t \geq 2$ vertices (the case $t = 1$ is obvious). Let xy be an edge of T where y is a leaf and put $S = T - y$. Our aim is to show that $e(G) \leq (t-2)n/2$, which follows from estimating $|\mathcal{B}_G(S, x)|$ in two ways.

Given $(\sigma, v_j) \in \mathcal{B}_G(S, x)$, the first vertex of σ can be any $u \in V(G)$, which gives $(n-1)!$ choices for completing the remaining $n-1$ vertices of σ. Moreover, since T is not contained in G, the vertex v_j in (σ, v_j) can be any neighbour of u. Therefore,

$$|\mathcal{B}_G(S, x)| = \sum_{u \in V(G)} \left((n-1)!\, d_G(u)\right) = (n-1)!\, 2e(G). \tag{1.1}$$

Equality (1.1) together with the following claim applied to S and x gives $e(G) \leq (t-2)n/2$.

Claim 1.1. *For every rooted tree (T, x) we have $|\mathcal{B}_G(T, x)| \leq n!e(T)$.*

Proof of Claim 1.1. We use induction on $e(T)$. If $e(T) = 0$, then T contains only the vertex x and then $\mathcal{B}_G(T, x) = \varnothing$ as any subgraph of G with one vertex contains T. Assume $e(T) \geq 1$ and $x \in V(T)$. Suppose that for every rooted tree (T', z) with $z \in V(T')$ and $e(T') < e(T)$ we have

$$|\mathcal{B}_G(T', z)| \leq n!e(T'). \tag{1.2}$$

Let w be an arbitrary neighbour of x in T, and let T_x and T_w be the components of $T - \{xw\}$ containing x and w, respectively. To facilitate the counting, we split $\mathcal{B}_G(T, x)$ into three disjoint sets. Note that since every copy of (T, x) in G contains a copy of (T_x, x) with the same root, we have $\mathcal{B}_G(T_x, x) \subseteq \mathcal{B}_G(T, x)$. From (1.2), we have

$$|\mathcal{B}_G(T_x, x)| \leq n!e(T_x). \tag{1.3}$$

It remains to estimate $|\mathcal{B}_G(T, x) \setminus \mathcal{B}_G(T_x, x)|$. For each pair $(\sigma, v_j) \in \mathcal{B}_G(T, x) \setminus \mathcal{B}_G(T_x, x)$, let r be the smallest index such that $G[\{v_1, \dots, v_r\}]$ contains a copy of (T_x, x) rooted at v_1. Since the pair $(\sigma, v_j) \notin \mathcal{B}_G(T_x, x)$, we have $r < j$ because, from the definition of $\mathcal{B}_G(T_x, x)$, there is a copy of (T_x, x) rooted at v_1 in $G[\{v_1, \dots, v_{j-1}\}]$.

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It is convenient to define an injective map $(\sigma, v_j) \mapsto (\tau, v_1)$ that changes the order of some "blocks" of consecutive vertices of σ. This map is described below.

$$\sigma = (\boxed{v_1, \dots, v_r}\ \boxed{v_{r+1}, \dots, v_{j-1}}\ \boxed{v_j}\ \boxed{v_{j+1}, \dots, v_n});$$

$$\tau = (\boxed{v_j}\ \boxed{v_{r+1}, \dots, v_{j-1}}\ \boxed{v_1, \dots, v_r}\ \boxed{v_{j+1}, \dots, v_n}).$$

Note that this map in in fact injective: (τ, v_1) determines $\boxed{v_{r+1}, \dots, v_{j-1}}$ and $\boxed{v_j}$ in σ. Moreover, from the minimality of r, we recover $\boxed{v_1, \dots, v_r}$ as the shortest segment of τ starting at v_1 that contains a copy of (T_x, x) rooted at v_1. Finally, the final segment $\boxed{v_{j+1}, \dots, v_n}$ of σ is as in τ.

Since $(\sigma, v_j) \mapsto (\tau, v_1)$ is injective, we can bound $|\mathcal{B}_G(T, x) \setminus \mathcal{B}_G(T_x, x)|$ by counting the pairs $(\sigma, v_j) \in \mathcal{B}_G(T, x) \setminus \mathcal{B}_G(T_x, x)$ depending on whether the corresponding (τ, v_1) belongs to the set $\mathcal{B}_G(T_w, w)$. Denote by $\mathcal{B}_{\text{in}}$ the family of these pairs (σ, v_j) with $(\tau, v_1) \in \mathcal{B}_G(T_w, w)$, and by $\mathcal{B}_{\text{out}}$ the family of these pairs (σ, v_j) with $(\tau, v_1) \notin \mathcal{B}_G(T_w, w)$.

– *Pairs $(\sigma, v_j) \in \mathcal{B}_{in}$:* For any (σ, v_j) in $\mathcal{B}_{\text{in}}$, the associated pair (τ, v_1) belongs to $\mathcal{B}_G(T_w, w)$. Then, the subgraph $G[\{v_j, v_{r+1}, \dots, v_{j-1}\}]$ contains no copy of (T_w, w) rooted at v_j. Since $(\sigma, v_j) \mapsto (\tau, v_1)$ is injective, each pair in $\mathcal{B}_G(T_w, w)$ is associated with at most one pair (σ, v_j) in $\mathcal{B}_G(T, x) \setminus \mathcal{B}_G(T_x, x)$. Since $e(T_w) < e(T)$, the induction hypothesis (1.2) applies to (T_w, w). Therefore,

$$|\mathcal{B}_{\text{in}}| \le |\mathcal{B}_G(T_w, w)| \le n! e(T_w). \tag{1.4}$$

– *Pairs $(\sigma, v_j) \in \mathcal{B}_{out}$:* For $(\sigma, v_j) \in \mathcal{B}_{\text{out}}$, the pair (τ, v_1) is not in $\mathcal{B}_G(T_w, w)$. Then, since $v_1 v_j \in E(G)$, there is a copy of (T_w, w) rooted at v_j in $G[\{v_j, v_{r+1}, \dots, v_{j-1}\}]$. This copy of (T_w, w) together with the copy of (T_x, x) rooted at v_1 in $G[\{v_1, \dots, v_r\}]$ and the edge $v_1 v_j$ gives a copy of (T, x) rooted at v_1 in $G[\{v_1, \dots, v_j\}]$. Since $(\sigma, v_j) \in \mathcal{B}_G(T, x)$, no such copy exists before v_j. In view of this, v_j is the first vertex in σ whose inclusion produces a copy of (T, x) rooted at v_1. Then, for a fixed ordering σ, once this copy of (T, x) appears, it remains in every longer prefix of σ, which implies that no (σ, v_k) with $k > j$ belongs to $\mathcal{B}_G(T, x)$. From this, we conclude that for each one of the possible $n!$ orderings σ there is at most one pair (σ, v_j) in $\mathcal{B}_{\text{out}}$. Therefore,

$$|\mathcal{B}_{\text{out}}| \le n!\,. \tag{1.5}$$

Putting (1.3), (1.4) and (1.5) together, we obtain $|\mathcal{B}_G(T, x)| \le n!(e(T_x) + e(T_w) + 1) = n! e(T)$, which concludes the proof of the claim. □

Claim 1.1 applied to S and x gives $|\mathcal{B}_G(S, x)| \le n! e(S) = n!(t-2)$, which together with (1.1) implies $e(G) \le (t-2)n/2$, as required. □

2. Use of AI and purpose of this note

The ideas presented here were obtained by providing instructions to GPT-6 Astra to simplify the proof given in [2]. I worked on these ideas and wrote this note, which is intended to provide a simpler version of the proof of the Erdős–Sós Conjecture given in [2].

References

[1] P. Erdős, Extremal problems in graph theory, in *Theory of Graphs and its Applications* (Proc. Sympos. Smolenice, 1963), M. Fiedler (ed.), Publishing House of the Czechoslovak Academy of Sciences, Prague, 1964, pp. 29–36.

[2] *A Counting Proof for Erdős Problem 548*. Manuscript. https://www.erdosproblems.com/static/548-proof.pdf.

[3] T. Adamczewski and T. F. Bloom. *FrontierMath Erdős*. https://epoch.ai/files/frontiermath-erdos.pdf.

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